

Title

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Abstract—The abstract is a single paragraph of 150–250 words, written last, once results already exist. It must be an objective synthesis of the whole paper, without stating conclusions that are not supported in the body of the text. Even though it has no visible subheadings, draft it following this internal structure:

- (1) **Context:** what question or need motivates the work?
- (2) **Methods:** what was done to address it? (approach, materials, main methods, in 1-2 lines).
- (3) **Results:** what was found? (the main findings, with concrete data if possible).
- (4) **Conclusion:** what does that finding imply?

Avoid mathematical or chemical formulas inside the abstract.

Index Terms—keyword 1, keyword 2, keyword 3 (3 to 6 terms specific to the topic of the paper, yet reasonably common within the discipline – they help other researchers find the work).

I. INTRODUCTION

THE introduction should place the work in a broad context and explain why it matters. It should cover, in this order:

- The problem or question that motivates the work, explained so that someone outside the specific field can understand it.
- The **motivation**: why this problem matters, who is affected by it, and what happens if it is not addressed.
- A brief preview that related work is reviewed next (the actual review goes in the Related Work subsection below).
- The gap, limitation, or open controversy that the reviewed related work leaves unresolved, and that this paper addresses.
- The concrete objective of the work.
- A preview of the main conclusion (one or two sentences).

Citation example: citing a source [1]. To cite several sources at once: [2], [3]. A closing paragraph that “summarizes the structure of the paper” is not required unless your journal/course explicitly asks for it.

A. Related Work

Recent advances in real-time computer vision and pose tracking leverage lightweight deep learning frameworks. Quiñonez *et al.* [1] demonstrated the application of Machine Learning solutions with MediaPipe for real-time landmark estimation. Technical tutorials and implementations using MediaPipe 101 [2] and OpenCV [3] provide foundational structures for real-time object detection and feature extraction [4]. Building upon these computer vision solutions, this work integrates 21-point hand tracking and body pose estimation directly into the kinematic control of the SOFIA active surface system.

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II. MATERIALS AND METHODS

Table I summarizes the bill of materials for the SOFIA 16-module system, including each component’s technical description, manufacturer, and country of origin.

The servomotor operational limits and step response parameters were experimentally evaluated using the test protocol recorded in “docs/experimental-data/servomotor_pruebas_caracterizacion.xlsx”. Under continuous PWM excitation at 50 Hz, the effective angular deflection spanned $\theta \in [19.3^\circ, 70.2^\circ]$ with a mean operating angle of 47.9° and step response times ranging between 75 ms and 180 ms.

This section must have enough detail for someone else to replicate the work. Include, as applicable:

- The materials, components, equipment, datasets, or tools used, with relevant specifications (well-established methods do not need to be justified in detail; citing them is enough).
- The methods or procedures applied, described with enough detail to be reproducible. If a method is new, describe it in detail; if it is a known method, a citation plus a brief description is enough.
- If the work uses public data or generates its own dataset, state where it is available (or state that it will be provided during review).
- If the work involves human or animal subjects, or requires ethical approval, state the authority that granted it and the approval code. If not applicable, this can be omitted.
- If generative artificial intelligence was used (beyond superficial text editing) to generate text, data, or graphics, or to support the study design, data collection, or analysis, disclose it briefly here.

Example of a long quotation or of a highlighted note within the text (use this “quote” block sparingly).

III. DESIGN AND IMPLEMENTATION OF THE TECHNOLOGICAL SOLUTION

This section shows how the methods and materials from the previous section were turned into the actual built solution. Include, as applicable:

- The overall system architecture (block diagram) showing how the different subsystems interact.
- Mechanical, electronic, and/or software design decisions, with enough detail to be reproducible.
- How the different subsystems were integrated into the complete solution.
- Configuration, calibration, or programming steps needed to bring the system into working operation.

TABLE I
BILL OF MATERIALS SUMMARY FOR THE SOFIA 16-MODULE SYSTEM

Component	Qty.	Technical description	Manufacturer	Country / Location
MG90S Servomotor	32	Metal gear micro servo, rated torque 1.8–2.2 kgf-cm at 4.8–6.0 V, evaluated angle range $\theta \in [19.3^\circ, 70.2^\circ]$, 50 Hz PWM control	TowerPro	Shenzhen, China
Module Enclosure	16	3D-printed PLA enclosure, custom SOFIA003 honeycomb design for 2-DOF module	In-house (LabDST)	Mexico City, Mexico
M3 Fasteners (Various)	64	Stainless steel socket head cap screw, M3 $\times$ 10 mm, for servo and frame mounting	Generic	Mexico City, Mexico
PCA9685 Servo Driver	2	16-Channel 12-bit PWM I ² C driver module for 32 servomotors	NXP Semiconductors / Adafruit	Eindhoven, Netherlands
ESP32-C3 Master MCU	1	32-bit RISC-V microcontroller, 2.4 GHz Wi-Fi / Bluetooth 5 LE for trajectory generation	Espressif Systems	Shanghai, China

TABLE II
ESP32-C3 TECHNICAL SPECIFICATIONS AND COMPLIANCE ASSESSMENT.

Component	Parameter	Value	Source
ESP32-C3 (Espressif Systems, v2.4)	Supply Voltage (VDDA, VDD3P3, VDD3P3_RTC)	3.0–3.6 V; 3.3 V typical	Datasheet v2.4, Section 5.2, Table 5-2, p. 54
ESP32-C3	Cumulative Supply Current (IVDD)	0.5 A min.	Datasheet v2.4, Section 5.2, Table 5-2, p. 54
ESP32-C3	Peak Wi-Fi Transmission Current	276–335 mA, depending on the operating mode	Datasheet v2.4, Section 5.6.1, Table 5-7, p. 56
ESP32-C3	VIH: High-Level Input Voltage	$0.75 \times VDD$ min.	Datasheet v2.4, Section 5.4, Table 5-4, p. 55
ESP32-C3	VIL: Low-Level Input Voltage	-0.3 to $0.25 \times VDD$ V	Datasheet v2.4, Section 5.4, Table 5-4, p. 55
ESP32-C3	Strapping Pins / Boot Configuration Pins	GPIO2, GPIO8, and GPIO9. Their logic levels are sampled during reset to determine the boot mode.	Datasheet v2.4, Section 3, Boot Configuration, p. 30
ESP32-C3	GPIO8 and GPIO9 Boot-Time Levels	SPI Boot: GPIO9 = 1. Joint Download Boot: GPIO8 = 1 and GPIO9 = 0. The GPIO8 = 0, GPIO9 = 0 combination is not listed as a valid boot mode in Table 3-3.	Datasheet v2.4, Section 3.1, Table 3-3, p. 31
ESP32-C3	Wi-Fi	2.4 GHz, IEEE 802.11 b/g/n	Datasheet v2.4, Features section, p. 3
ESP32-C3	Bluetooth	Bluetooth 5	Datasheet v2.4, Features section, p. 3
ESP32-C3	Operating Temperature	56 °C	Datasheet v2.4, Section 1.2, Table 1-1, p. 12

TABLE III
SERVOMOTOR MG90S TECHNICAL SPECIFICATIONS AND COMPLIANCE ASSESSMENT.

Component	Parameter	Value	Source
MG90s	Supply and operating Voltage	4.8 V – 6.0 V	MG90S datasheet, Tower Pro, p.2
MG90s	Stall Torque	1.8 kgf-cm at 4.8V, 2.2 kgf-cm at 6V	MG90S datasheet, Tower Pro, p.1
MG90s	Operating Speed	0.1 s/60° at 4.8V, 0.08 s/60° at 6V	MG90S datasheet, Tower Pro, p.1
MG90s	Control Signal	PWM, orange wire; Vcc red; GND brown; 20 ms (50 Hz) period, 1–2 ms duty cycle	MG90S datasheet, Tower Pro, p.2
MG90s	Rotation Range	approximately 180 degrees (90° in each direction)	MG90S datasheet, Tower Pro, p.2
MG90s	Evaluated Angle Range (Experimental)	19.3° – 70.2° (mean: 47.9°)	LabDST test log (docs/experimental-data)
MG90s	Response Time (Experimental)	75 – 180 ms	LabDST test log (docs/experimental-data)
MG90s	Dead Band Width	5 μ s	MG90S datasheet, Tower Pro, p.2
MG90s	Operating Temperature	Not specified in datasheet	N/A
MG90s	Dimension & Weight	22.5 $\times$ 12 $\times$ 35.5 mm approx.; 13.4 g	MG90S datasheet, Tower Pro, p.1

Use figures (block diagrams, CAD renders, architecture diagrams, photographs of the assembled system) and tables to support this description, following the same figure/table conventions shown in the Results section below.

IV. RESULTS

A. Kinematic Tracking Simulation

To validate the analytical kinematic model, a dynamic tracking simulation of the 16-module SOFIA robotic honeycomb array was conducted in a virtual 3D environment using

the Unity game engine. The objective was to evaluate the coordinated orientation of the modules toward a moving target coordinate $\mathbf{p}(t) \in \mathbb{R}^3$.

For each module m_i , the inverse kinematics procedure determines the servo variables (q_1, q_2) from the relative position of the target with respect to the module base frame. The resulting joint angles are constrained according to the permissible actuator range. The complete inverse kinematics procedure is summarized in Algorithm 1.

TABLE IV
PCA9685 TECHNICAL SPECIFICATIONS AND COMPLIANCE ASSESSMENT.

Component	Parameter	Value	Source
PCA9685 (NXP Semiconductors, datasheet Rev. 4)	Supply Voltage (V_{DD})	2.3–5.5 V	Datasheet Rev. 4, Section 2, Table 14, p. 38
PCA9685	PWM Resolution	12 bits (4096 steps)	Datasheet Rev. 4, Section 1, p. 1
PCA9685	Number of PWM Channels	16 independent channels	Datasheet Rev. 4, Section 1, p. 1
PCA9685	Communication Interface	I ² C Fast-mode Plus	Datasheet Rev. 4, Section 2, p. 2
PCA9685	Maximum I ² C Frequency	Up to 1 MHz	Datasheet Rev. 4, Section 2, p. 2
PCA9685	PWM Frequency Range	24 Hz – 1526 Hz	Datasheet Rev. 4, Section 7.3.5, p. 25
PCA9685	Default PWM Frequency	Approximately 200 Hz	Datasheet Rev. 4, Table 8, p. 25
PCA9685	I ² C Address Configuration	A0–A5 pins, up to 62 devices	Datasheet Rev. 4, Section 7.1, p. 7
PCA9685	Output Current	25 mA per output	Datasheet Rev. 4, Table 13, p. 38
PCA9685	Total Output Current	400 mA maximum	Datasheet Rev. 4, Table 13, p. 38
PCA9685 Module (UNIT Electronics)	Output Ports	16 PWM outputs with V+, GND and PWM signal	UNIT Electronics product specification
PCA9685 Module (UNIT Electronics)	Logic Voltage	3–5 V	UNIT Electronics product specification
PCA9685 Module (UNIT Electronics)	Servo Power Supply (V+)	External supply power for connected actuators	UNIT Electronics product specification
PCA9685	Operating Temperature	−40 °C to +85 °C	Datasheet Rev. 4, Table 13, p. 38

Algorithm 1 SOFIA Module Inverse Kinematics

Require: Module base origin $\mathbf{r}_{0,2}$, target coordinate $\mathbf{p}$, joint angle limits $[q_{1,\min}, q_{1,\max}]$ and $[q_{2,\min}, q_{2,\max}]$

Ensure: Joint angles q_1, q_2

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1:  $\mathbf{r}_{2,p} \leftarrow \mathbf{p} - \mathbf{r}_{0,2}$ 
2:  $[u_x, u_y, u_z]^\top \leftarrow \mathbf{r}_{2,p} / \|\mathbf{r}_{2,p}\|$ 
3:  $q_2 \leftarrow \arcsin(u_x)$ 
4:  $q_1 \leftarrow \operatorname{atan2}(-u_y, u_z)$ 
5: if  $q_1 < q_{1,\min}$  then
6:    $q_1 \leftarrow q_{1,\min}$ 
7: else if  $q_1 > q_{1,\max}$  then
8:    $q_1 \leftarrow q_{1,\max}$ 
9: end if
10: if  $q_2 < q_{2,\min}$  then
11:    $q_2 \leftarrow q_{2,\min}$ 
12: else if  $q_2 > q_{2,\max}$  then
13:    $q_2 \leftarrow q_{2,\max}$ 
14: end if
15: return  $q_1, q_2$ 

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Figure 1 shows six representative frames of the simulated tracking process. As the target moves through the workspace, the modules progressively change their orientation, maintaining the direction of their normal vectors $\hat{\mathbf{r}}_{2,3}^{(i)}$ toward the target position. The sequence illustrates the coordinated response

of the 16-module array during different stages of the target trajectory.

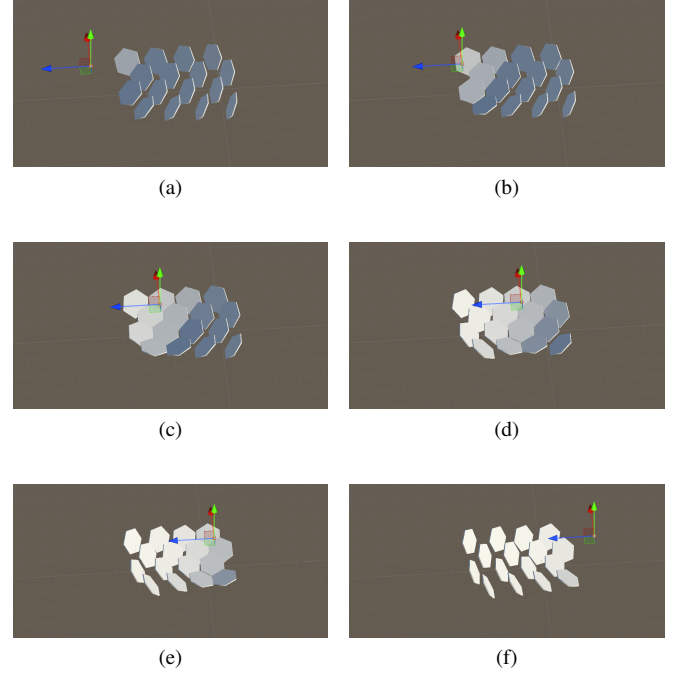


Fig. 1. Representative frames of the SOFIA kinematic tracking simulation. The six frames illustrate the progressive reorientation of the 16-module honeycomb array as the target coordinate $\mathbf{p}(t)$ moves through the workspace, with the module normal vectors $\hat{\mathbf{r}}_{2,3}^{(i)}$ oriented toward the target.

V. DISCUSSION

This is where the results are interpreted, not just repeated. It should answer:

- What do the obtained results mean?
- How do they compare with the state of the art cited in the introduction? Do they confirm, contradict, or qualify previous work?
- What limitations does the work have (methodological, of scope, of validity)?
- What practical or theoretical implications does the finding have?
- What future work does this open up?

VI. CONCLUSIONS

A brief synthesis (not a repetition of the abstract or of the discussion) of the answer to the question posed in the introduction, and of the concrete contribution of the work. This section is not mandatory if the discussion already provides that closure, but it is recommended when the paper is long or the discussion covered several topics.

AUTHOR CONTRIBUTIONS

If the paper has several authors, briefly specify each person's individual contribution (e.g., conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, writing, visualization, supervision, project administration, funding acquisition). Authorship must be limited to those who contributed substantially to the reported work.

DATA AVAILABILITY

State where the data, code, or materials that support the reported results can be found (repository, link, DOI). If no new data were generated, or if there are privacy or confidentiality restrictions, state that as well.

ACKNOWLEDGMENT

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CONFLICTS OF INTEREST

Choose whichever of the following applies, or adapt it to your case:

- **No conflict:** state “The authors declare no conflicts of interest.”
- **Declared conflict:** identify and declare any personal circumstances or interests that could be perceived as inappropriately

influencing the representation or interpretation of the reported results.

- **Role of funders:** state the role of any funders in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results. If there was no such role, state: “The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.”

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